Generalized Morse theory for distance functions

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1 Generalities on Lipschitz Functions

1.1 Lipschitz functions on manifolds and subdifferentials

For all the following, M will denote a differentiable manifold of dimension m. In this section, we generalize many definitions related to Lipschitz functions. We begin by extending the definition of Lipschitz functions to functions on a manifold.

Definition 1. *A function $f : M \to \mathbb{R}^n$, where M is a smooth manifold, is locally Lipschitz if for any chart $U \subset M$ and any compact $K \subset U$, there exists $C > 0$ such that, with the coordinates on U*

$$\forall x, y \in K, |f(x) - f(y)| \leq C|x - y|$$

Let $f : U \subset \mathbb{R}^m \to \mathbb{R}^n$ be a Lipschitz function, where U is open. Rademacher's theorem, which states that f is differentiable almost everywhere, will allow us to generalize differentials to this setting.

Definition 2. *[3] The subdifferential of f at x, denoted by $\partial_x f$ is the convex hull of the linear maps l obtained as*

$$l = \lim_{i \to \infty} d_{x_i} f$$

where f is differentiable at each x_i and $x_i \to x$.

The subdifferential is then set valued. This set is always non empty and compact, f being Lipschitz bounds df where it is defined.
By [2], we can exclude a set of measure 0 from the possible values of x_i without changing the definition.

Proposition 1. *∂f is the smallest convex valued upper semi-continuous function containing df whenever it is defined.*

Proof. First, ∂f is upper semi-continuous : by compactness, is suffices to show that for all $x \in U, \varepsilon > 0$, for y sufficiently close to x.

$$\partial_y f \subset \partial_x f + B(0, \varepsilon)$$

Assume for contradiction that there exists a sequence $x_i \to x$ such that for all i, $\partial_{x_i} f$ has an element out of $A := \partial_x f + \overline{B}(0, \varepsilon/2)$. A being convex and closed, let x_i' be such that $d_{x_i'} f$ is out of A, and such that $x_i' - x_i \to 0$. Up to extraction, $d_{x_i'} f$ converges to an element in $\partial_x f$, so it enters A, contradiction.
The second part follow from ∂f being a convex hull. □

The subdifferential allows us to generalize the idea of regular and critical points, which will be an important tool for generalizing Morse theory. The definition is as follows :

Definition 3. *A point $x \in U$ will be called critical if there exists some $l \in \partial_x f$ not of maximal rank. x will be called regular if it isn't critical.*

To extend these subdifferentiability notions to manifolds, one must guarantee that for all charts, the definitions of ∂f coincide. In this is the case, $\partial_x f$ is a compact convex subset of $T_x^* M$. The following generalization of the chain rule allows us to do that.

Proposition 2 (Subdifferential chain rule)**.** *Let $h : \mathbb{R}^n \to \mathbb{R}^k$ be a C^1 function. For $x \in U$:*

$$\partial_x (h \circ f) \subset d_{f(x)} h \circ \partial_x f$$

If h is a diffeomorpism, we have equality.

Proof. For the first part, because f is differentiable a.e, it suffices to show that the leftwise term is an upper semi-continuous convex valued function that contains $d(h \circ f)$ whenever df is defined. This follows from the definition of ∂f.

If h is a diffeomorphism, then for $x \in U$:

$$\partial_x f = \partial_x (h^{-1} \circ h \circ f) \subset d_{h \circ f(x)} h^{-1} \circ \partial_x (h \circ f) \subset \partial_x f$$

so we have equalities, and the result follows. □

Although some important differentiablity related results can be obtained with locally Lipschitz functions as we'll see, a general locally Lipschitz function can have some pathological behaviours. As an example, it's subdifferential isn't in general a topological submanifold of M as illustrated by Figure ??. However, the functions we'll study, for example continuous selections (see Section 2) have much nicer behaviour and we retrieve this property.

1.2 Generalized differential theorems

We now consider the specific case of $f : U \subset \mathbb{R}^m \to \mathbb{R}^m$. This section is devoted to proving generalizations by Clarke [3] of the inverse function theorem and the implicit function theorem for the case of Lipschitz functions.

In this section, we norm the space of linear maps $\mathbb{R}^m \to \mathbb{R}^m$, which we denote $\mathcal{L}(\mathbb{R}^n)$, with the operator norm $\|\cdot\|$ obtained from the norm used on $\mathbb{R}^n$.

Lemma 1. *Assume U is convex, and that there exists some closed convex set C such that*

- *Every $l \in C$ is invertible.*
- *$\partial_x f \subset C$ for all $x \in U$*

Then there exists some $\delta > 0$ such that for all $x, y \in U$:

$$|f(x) - f(y)| \geq \delta |x - y|$$

Proof. Firstly, we set

$$\delta = \inf \{|l(h)| \mid l \in C, h \in S\}$$

where S is the unit sphere in $\mathbb{R}^n$. $\delta > 0$ because all $l \in C$ are invertible and $C \times S$ is closed.

Let $x, y \in U$ and assume $x \neq y$. Let π be the plane perpendicular to $y - x$ passing through x. f being differentiable a.e, Fubini's theorem implies that for almost all $x' \in \pi$, f is differentiable almost on the whole line passing through x' and perpendicular to π (i.e f isn't differentiable on a subset of this line of one dimensional measure 0).
Denoting $F : t \longmapsto f(x' + t(y - x))$, we have :

$$f(x' + y - x) - f(x') = \int_0^1 F'(t)dt = \int_0^1 d_{x'+t(y-x)} f(y - x)dt \tag{1}$$

Define the linear map

$$L = \int_0^1 d_{x'+t(y-x)} f dt$$

C being convex and closed, $L \in C$. Observe that the integral on the right of (1) is just $L(y - x)$. So

$$|f(x' + y - x) - f(x')| \geq \delta |y - x|$$

This inequality holds for almost all $x' \in \pi$, so by continuity of f, it holds for x and we get the desired result. □

Theorem 1 (Lipschitz inverse function theorem)**.** *If x_0 is a regular point of f, then there exists V, W, neighbourhoods of x_0 and $f(x_0)$ respectively, such that f induces a bijection $f : V \to W$ where f^{-1} is Lipschitz too.*

Proof. By compactness of $\partial_{x_0} f$, regularity of x_0, and the fact that the invertible linear maps form an open set, there exist some $\varepsilon > 0$ such that the set

$$C := \partial_{x_0} f + \overline{B}(0, \varepsilon)$$

is composed only of invertible maps. By upper semi-continuity of ∂f, there is some $r > 0$ such that for $y \in B(x_0, r)$, $\partial f(y) \subset C$. We then apply Lemma 1 with $U = B(x_0, r)$ and get a corresponding $\delta > 0$.

Set $V = B(x_0, r/2)$ and $W = f(V)$. We showed in particular that f is injective on $\overline{V}$, by compactness $f : \overline{V} \to f(\overline{V})$ induces a homeomorphism. W is then open. The inequality we got with δ implies that f^{-1} is $\frac{1}{\delta}$ Lipschitz. $\square$

From this theorem, follows the implicit function theorem for Lipschitz functions

Theorem 2. *Let $F : U \to \mathbb{R}^n$, where $U \subset \mathbb{R}^m \times \mathbb{R}^n$ is open and F is Lipschitz. Let $(x_0, y_0) \in U$ such that $F(x_0, y_0) = 0$ and y_0 is a regular point of $y \mapsto F(x_0, y)$. Then there exist neighbourhoods $(x_0, y_0) \in W \subset U$, $x_0 \in V \subset R^m$ and a Lipschitz function $g : V \to \mathbb{R}^n$ such that*

$$(x, y) \in W \text{ and } F(x, y) = 0 \Leftrightarrow x \in V \text{ and } y = g(x)$$

The proof is almost the same as the usual implicit function theorem, where the continuity of the differential is replaced by the semi continuity of ∂F.

1.3 First Morse lemma for locally Lipschitz functions

The goal of this section is to use the generalized notion of critical points in order to generalize the first Morse lemma to the case of locally Lipschitz functions. Fix M an m-dimensional smooth manifold, $f : M \to \mathbb{R}$ a continuous function and define for all $c \in \mathbb{R}$ the sublevel set of f of value c as

$$M_c := f^{-1}(]-\infty, c])$$

The classical first Morse lemma states that if f is C^1, and there are no critical points in $f^{-1}([a, b])$ (that we assume to be compact), then M_b is diffeomorphic to M_a and M_a is a deformation retract of M_b. We extend this to locally Lipschitz functions, following [1]. Note that in this setting x is a critical point of f iff $0 \in \partial_x f \subset T_x^* M$.

Proposition 3. *Let $a < b \in \mathbb{R}$ such that $f^{-1}([a, b])$ is compact and free of critical points. Then M_a is Lipschitz homeomorphic to M_b, M_a is a deformation retract of M_b, the homotopy being locally Lipschitz.*

Proof. We'll construct $V \in \mathrm{Vec}(M)$ a smooth complete vector field on M such that, for all $x \in M$ such that $d_x f$ is defined :

$$d_x f(V(x)) \leq 0 \text{ and } d_x f(V(x)) \leq -1 \text{ if } x \in f^{-1}([a, b]) \tag{2}$$

Let $x \in f^{-1}([a, b])$. x being regular and $\partial_x f$ convex, we find an affine hyperplane in $T_x^* M$ separating it from 0. This gives us $v_x \in T_x M$ such that $\langle \partial_x f, v_x \rangle \leq -2$. By semi continuity, there exists an open neighbourhood U_x of x such that for all $y \in U_x$, $\langle \partial_y f, v_x \rangle \leq -1$. We choose this neighbourhood such that it has compact closure.

These U_x cover $f^{-1}([a, b])$. Let $U_{x_1}, ..., U_{x_r}$ be a finite subcover, and let $\rho_1, ..., \rho_r$ be a partition of unity subordinated to this covering. We set

$$\forall x \in M, V(x) = \sum_{i=1}^{r} \rho_i(x) v_{x_i}$$

which indeed verifies (2) : if $d_x f$ is defined, it is contained in $\partial_x f$, and V is of compact support so it is complete.

Now observe that for $x \in f^{-1}([a, b])$, $t \mapsto f(e^{tV} x)$ is strictly decreasing. Denote by $t(x, s)$, $s \in [0, 1]$, the unique time such that $f(e^{t(x,s)V} x) = as + f(x)(1 - s)$. The equation defining $t(x, s)$ is locally Lipschitz in s, x and t, so by the Lipschitz implicit function theorem, t is a locally Lipschitz function of (x, s). The following homotopy verifies what we want

$$H : \begin{array}{ccc} [0,1] \times M_b & \longrightarrow & M_b \\ (s, x) & \longmapsto & \begin{cases} e^{t(x,s)V} x & \text{if } x \in f^{-1}([a, b]) \\ x & \text{else} \end{cases} \end{array}$$

We can similarly define t' a locally Lipschitz function of x such that $f(e^{t'(x)V}x) = b$ for x in the orbit of some $y \in f^{-1}(b)$ by V. Denote O the set of $x \in M$ that are in such orbits. The homeomorphism is then given by

$$\begin{array}{rcl} M_b & \longrightarrow & M_a \\ x & \longmapsto & \begin{cases} e^{t(e^{t'(x)V}x,1)V}x & \text{if } x \in O \\ x & \text{else} \end{cases} \end{array}$$

□

2 Morse theory of continuous selections

2.1 Definitions

We now restrict ourselves to continuous functions that are piecewise smooth with finitely many pieces. Following (ref), we define those as continuous selections.

Definition 4. *A continuous function $f : M \to \mathbb{R}$ is a continuous selection if there exists $\varphi_1, ..., \varphi_n$ "regular" functions such that*

$$\forall x \in M, f(x) \in \{\varphi_1(x), ..., \varphi_n(x)\}$$

The term "regular" is used loosely here to indicate that the φ_i satisfy some extra condition on top of being continuous. For what follows, we'll restrict ourselves to continuous selections of smooth functions (but C^2 will suffice).

Note that a continous selection of locally Lipschitz functions is itself locally Lipschitz : if M is an open set of $\mathbb{R}^m$, and we assume that the functions φ_i are Lipschitz with constants c_i, then f is Lipschitz with constant $\max_i c_i$. Because smooth functions are locally Lipschitz, the functions we'll study will be too.

We can then relate ∂f to the differentials of the φ_i. Indeed, a limit of points of the form $d_x\varphi_i$ in T^*M is of the same form. Define the set of active indices at x,

$$I(x) = \{i \in [\![1, n]\!] \mid \varphi_i(x) = f(x)\}$$

$\partial_x f$ is then the convex hull of some $d_x\varphi_i$, $i \in I(x)$. It is possible that not all $d_x\varphi_i$ are used, if for example one of the functions isn't active in a neighbourhood of x.

We would like to define what it means for such a continuous selection to be Morse. Let f be a continuous selection of $\varphi_1, ..., \varphi_n$. Following [1], we give three conditions for a function to be considered Morse.

$$\text{For all } i \neq j, \text{ the graphs of } \varphi_i \text{ and } \varphi_j \text{ are transversal.} \tag{C1}$$

This condition has some important consequences : for a non empty $I \subset [\![1, n]\!]$, we define $W_I := \{x \in M \mid I(x) = I\}$. Condition (C1) implies that W_I is a submanifold of M of codimension $|I| - 1$: the intersection of the graphs of $\varphi_i, i \in I$ is a submanifold of $M \times \mathbb{R}$ of codimension $|I|$ and W_I is the pre-image of this intersection through the diffeomorphism $x \mapsto (x, \varphi_i(x))$, $i \in I$, with the codomain being the graph of φ_i.

This transversality condition also implies that for $x \in M$, the differentials $d_x\varphi_i$ are affinely independant where $i \in I(x)$. In particular, $\partial_x f$ is a simplex with less than $|I(x)| - 1$ vertices.

Note that for all $I \subset [\![1, n]\!]$, $f_{|W_I}$ is a smooth function. This motivates the second condition

$$\text{For all non empty } I \subset [\![1, n]\!], f_{|W_I} \text{ is Morse} \tag{C2}$$

Consider $x \in M$ a critical point of f. Observe that x must also be a critical point of $f_{|W_{I(x)}}$. So $T_xW_{I(x)} \subset \bigcap_{i \in I(x)} d_x\varphi_i$, but the converse can fail if the $d_x\varphi_i$ are not linearly independant enough. The last condition fixes that

$$\forall x \in M, \forall j \in I(x), (d_x\varphi_i)_{\substack{i \in I(x) \\ i \neq j}} \text{ are linearly independant} \tag{C3}$$

Geometrically, this condition corresponds to requiring that 0 is either outside or in the interior of the simplex $\mathrm{conv}\{d_x\varphi_i \mid i \in I(x)\}$, but not on one of its faces. Note (i think ??) that this condition can be relaxed by restricting it to the subset of $I(x)$ participating in $\partial_x f$.

A continuous selection f is then called a Morse function if it verifies all three conditions. Note that as in the smooth case, a generic continuous selection is Morse : (C1) is a generic condition, any two functions can be slightly perturbed in order for their graphs to be transversal, and any small perturbation of transversal graphs preserves transversality. By genericity of smooth Morse functions, (C2) is also a generic condition. With the geometric interpretation, we can also see that condition (C3) is generic.

2.2 Normal form of a Morse continuous selection

We show here an equivalent of the second Morse lemma for the case of Morse continuous selections. Noticing that $x \mapsto I(x)$ is upper semi-continuous and using the definitions yields the following result.

Lemma 2. *If $f : M \to \mathbb{R}$ is a Morse continuous selection, all its critical points are isolated.*

We'll then consider, up to restricting ourselves to a small enough chart, that $f : \mathbb{R}^m \to \mathbb{R}$ is Morse and has no critical points other than 0. We'll write for shorthand $W := W_{I(0)}$, $k := |I(0)| - 1$, $c := f(0)$, and assume that $I(0) = [\![1, k+1]\!]$.

Theorem 3. *If 0 is a critical point of f, then there exists Lipschitz local coordinates around 0, $\psi : \mathbb{R}^k \times \mathbb{R}^{m-k} \to U$ such that*

$$\forall y \in \mathbb{R}^k, \forall z \in \mathbb{R}^{m-k}, f(\psi(y,z)) = \lambda(y) + q(z)$$

where λ is a continuous selection of $k+1$ linear functions, and q is of the form

$$\forall z \in \mathbb{R}^{m-k}, q(z) = c - \sum_{i=1}^{\mu} z_i^2 + \sum_{i=\mu+1}^{m-k} z_i^2$$

with $0 \leq \mu \leq m-k$.

Proof. We first restrict ourselves to O a neighbourhood of 0 such that W can be considered rectified in some smooth coordinates i.e, $W \cap O = 0 \times \mathbb{R}^{m-k}$. We can also use the smooth Morse lemma to get $f(0,z) = q(z)$. For $(y,z) \in \mathbb{R}^m$, write $f^0(y,z) = f(y,z) - f(0,z)$. We will build for all z close enough to 0 a locally invertible map $\psi_z : \mathbb{R}^k \to \mathbb{R}^k$ such that $\psi_z(0) = 0$ and for y close enough to 0, $f^0(\psi_z(y), z)$ is a continuous selection of linear maps independant of z.

For what follows we'll fix z and omit it in the notations for convenience. We find a neighbourhood U of $0 \in \mathbb{R}^k$ such that

- U is connected.
- For all $y \in U$, $I(y) \subset [\![1, k+1]\!]$.
- For all $j \in [\![1, k+1]\!]$, the set O_j of points in U such that $f(y) \neq \varphi_j(y)$ is open.
- There exists $\psi_j : U \to \mathbb{R}^k$ a diffeomorphism such that for all $i \neq j$, $\varphi_i(\psi_j(y)) = \varphi_i(0) + d_0\varphi_i(y)$
- For all $y \neq 0$, $\psi_1(y), ..., \psi_{k+1}(y)$ are affinely independant.

We now want to combine these to take account of the possible intersections between the O_j. Our goal is to combine to ψ_j together with a partition of unity $\mu_j : U \backslash \{0\} \to \mathbb{R}$ and set our final diffeomorphism to be

$$\psi(y) = \sum_{i=1}^{k+1} \mu_i(y)\psi_i(y)$$

Moreover, ψ should take all active φ_i to its affine approximation.
Let $\Omega \subset U$ be a maximal set on which this is possible. Assume there exists some non zero $y \in \Omega$ verifying that all of these exist at y. Let j be such that $y \in O_j$. Set

$$F : \begin{array}{ccc} U \times \mathbb{R}^k_{<1} & \longrightarrow & \mathbb{R}^k \\ (y, (\mu_i)_{i \neq j}) & \longmapsto & \left(\varphi_i \left(\sum_{k \neq j} \mu_k \psi_k(y) + \left(1 - \sum_{k \neq j} \mu_k \right) \psi_j(y) \right) - \varphi_i(0) - d_0\varphi_i(y) \right)_{i \neq j} \end{array}$$

By assumption we have $F(y, (\mu_i(y)_{i \neq j})) = 0$. We can choose the ψ_j to be affinely independant because the $d_0\varphi_i$ are. Then, we use linear independance of the remaining k $d_0\varphi_i$ to show that the differential of F along the μ coordinates is invertible. So by the implicit function theorem, there exists a neighbourhood of y on which μ_i $i \neq j$ can be extended, and because they sum up to 1, we can also define μ_j. So Ω is open in $U \backslash \{0\}$.

Ω is also closed in this set : assume there is a sequence (y_r) in Ω converging to some non zero $y \in O_j$ for some j. By construction, the μ_i are bounded, so we can assume up to extraction that $\mu(y_r) \to \tilde{\mu}$. Then $F(y, (\tilde{\mu}_i)_{i \neq j}) = 0$, and we again use the implicit function theorem to build functions $\tilde{\mu}_i$ on a ball around y. Now take any point $y' \in \Omega$ in this ball. The implicit function theorem states that the μ_i are the only functions that gives 0 when fed into F and defined on a certain ball around y'. $\tilde{\mu}_i$ also satisfies these conditions, so $\tilde{\mu}_i(y') = \mu_i(y')$, so we can extend μ_i to y, $y \in \Omega$.

Because $U \backslash \{0\}$ is connected, so $\Omega = U \backslash \{0\}$ because we can trivially define μ_i at a point where only one index isn't active. ψ is then a local diffeomorphism using the fact that $d_0\varphi_i \circ d_0\psi = d_0\varphi_i$ for all i.

We constructed for all z close enough to 0 to approximate $d_{(0,z)}\varphi_i$ by $d_0\varphi_i$, ψ_z. They verify the following locally Lipschitz equation : $f(\psi_z(y), z) = \lambda(y) + q(z)$ where λ is a continuous selection of $d_0\varphi_i$. So $(z, y) \mapsto \psi_z(y)$ is locally Lipschitz.

$\square$

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